Scattering Transformation is a Neural Network

Ho Duc Tho and Takanori Miyoshi

Abstract— **In this paper, it will be demonstrated that the classical scattering transformation is in fact a neural network. This insight is helpful in the sense that it connects the classical scattering transformation theory with the neural network field. By using this principle, we can deductively employ the knowledge suggested by the neural network theory to design new scattering transformation-based communication laws that can resolve several technical problems prevailing in the telerobotics field. As an example, the total number of neurons of the classical scattering transformation is increased to enable independent contact force feedback to enhance teleoperation transparency. A simulation result is provided.**

I. Introduction

Classical scattering transformation was employed in [1] to passivate two-port communication channels subjected to constant time delays as found in the force-reflecting teleoperation architecture. Since this major breakthrough, a series of works were proposed to improve the classical scattering transformation [2]–[6]. Notably, by graphically interpreting the structure of the scattering transformation of [1], the notion of wave variable was introduced in [7], by which the scattering transformation can be employed to passivate any two-port system that has H_∞ norm less than one (i.e., it includes the delayed communication channel as a special case). Early researches utilizing the scattering transformation in teleoperations often followed the modular design of passivity formalism via passivity theorem. A more recent approach was to use the scattering transformation under the framework of small-gain theorem. This trend was ignited by the work of [8] with the introduction of a generalized scattering transformation applied on network control systems and later extended to planar conic systems [9] and non-planar conic ones [10]. Recently, an upper strictly passive scattering transformation was established to enable the use of direct contact force feedback (instead of coordinating force feedback) to enhance transparency of teleoperations [11].

In this paper, we will provide a different interpretation of the seminal work [1], i.e., it is argued that the classical scattering transformation is indeed a neural network. This new interpretation is simple but useful in the sense that it connects two (previously perceived as unrelated) concepts—the theory of classical scattering transformation and the neural network. In this light, various neural network-inspired scattering transformations can be developed to address some technical difficulties in the telerobotics field related to the time delay. One of the biggest strengths of a neural network rests on the fact that it can accept an unlimited amount of data and information if the total number of neurons (and weights) is increased accordingly. We can leverage this property to design a new scattering transformation, named force-receptive scattering transformation, whose total number of neurons is increased compared to the classical scattering transformation. The proposed force-receptive scattering transformation is capable of receiving both kinematics (positions and velocities of robots) and kinetics (true contact forces) information *independently* as inputs and modulating the outputs appropriately so that the communication channel subject to constant time delays become passive. This is the main contribution of the paper. Note that, stability of the closed-loop system will be conducted entirely on the basis of passivity formalism, as opposed to the approach based on the small-gain theorem as found in [9] and [10].

Notations: $\mathcal{L}_\infty$ denotes the space of uniformly bounded functions. The set of real numbers, non-negative real numbers, and non-negative integers are represented by $\mathbb{R}$, $\mathbb{R}^+$ and $\mathbb{N}$, respectively. The n-dimensional Euclidean space is $\mathbb{R}^n$. Moreover, let $\mathcal{N}$ denote the set $\{1, 2, 3\}$.

II. Scattering transformation is a neural network

The structure of a generalized scattering transformation is shown in Fig. 1, in which T is the magnitude of constant time delays (see [8] and [11]). It can be seen that each left and right scattering transformation consists of 2 neurons with 4 weights w_{ij} and v_{ij} $(i, j = 1, 2)$. The activation functions of all neurons are linear (or identity) and their biases are zero. It is observed that the structure of the scattering transformation closely resembles a neural network architecture called autoencoder [12, p. 493], which is shown in Fig. 2. By comparing Fig. 1 and Fig. 2, the left and right scattering transformations are in places of the encoder and decoder, respectively. In principle, an autoencoder is trained to copy its input to its output. However, it is not useful if autoencoders simply learn to produce a perfect copy. Alternatively, autoencoders are designed to imperfectly replicate their inputs, and to only retain desired and compressed attributes of data. The scattering transformation shares the same characteristic, i.e., the weights w_{ij} and v_{ij} are devised so that direct data transfer is prohibited while the desired passivity feature and steady-state transmission requirement are preserved in the presence of time delays [11]. Although the scattering transformation is very similar to a conventional autoencoder in both structure and functionality, they do have a fundamental difference, namely that, the conventional autoencoder is unilateral whereas the scattering transformation

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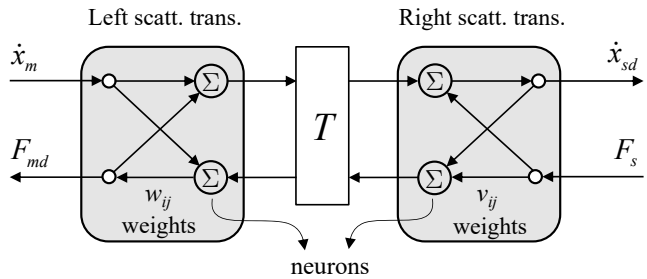

Fig. 1. Scattering transformation is a neural network.

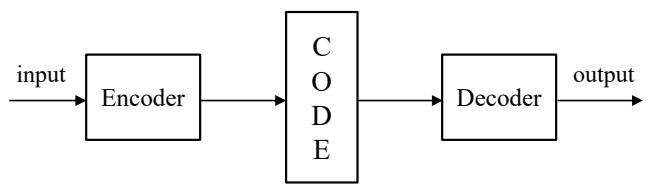

Fig. 2. The structure of a unilateral autoencoder.

is bilateral. More specifically, the left and right scattering transformations perform both encoding and decoding tasks. Thus, the generalized scattering transformation defines a new (bilateral) autoencoder (see Fig. 3).

To achieve better haptic transparency, a teleoperation system should be able to communicate two types of information between the local and remote sides: kinematics (positions and velocities of robots) and kinetics (actual contact forces). Provided that the classical scattering transformation has only 4 neurons (2 for each local and remote side), in order to transmit both kinematics and kinetics data over the network, one needs to mix them in the form of a weighted sum before submitting to the scattering transformation blocks as inputs (see [13]). Besides its unintuitive nature, such a technique poses a sinigficant difficulty in proving/ensuring stability for the closed-loop system. In the next section, we will design a force-receptive scattering transformation that can accept both kinematics and kinetics information *independently*, to passivate the communication channel subject to constant time delays while maintaining the integrity of the data transmission in the steady-state.

III. FORCE-RECEPTIVE SCATTERING TRANSFORMATION DESIGN AND ANALYSIS

To clearly explain the basic concept, this section is devoted to designing a force-receptive scattering transformation for a three-channel teleoperation system. The proposed scattering transformation framework for other teleoperation architectures with four or more channels can be adapted in the same manner. The force-receptive scattering transformation structure is shown in Fig. 4, where the total number of neurons and weights is increased to 6 and 18, respectively. All activation functions are remained linear and their biases are zero. The to-be-determined variables are the weights w_{ij} and v_{ij} $(i, j \in \mathcal{N})$. The local (or master) site sends a signal u_1 and receives the signals u_2 and u_3. The remote (or slave)

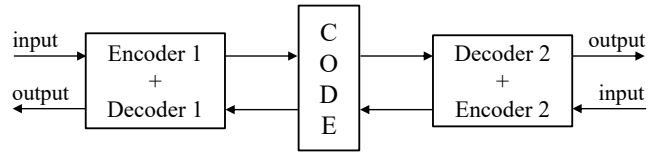

Fig. 3. A bilateral autoencoder.

site transmits the signals y_1 and y_3, and it also acquires a signal y_2. All sending and receiving signals are subjected to a constant time delay T, namely that,

$$\begin{aligned} z_{s1}(t) &= u_{m1}(t-T) \\ z_{m1}(t) &= u_{s1}(t-T) \\ z_{m2}(t) &= u_{s2}(t-T). \end{aligned} \tag{1}$$

The force-receptive scattering transformation needs to fulfill two requirements: passivity and steady-state transmission condition (see Definition 1). These requirements can be realized by devising proper neural weights w_{ij} and v_{ij}, as elaborated in the next subsections.

A. Passivity enforcement

To derive suitable constraints for the weights w_{ij} and v_{ij} so that the communication channel (orchestrated by the force-receptive scattering transformation) is passive, the following Lyapunov-Krasovskii storage function is chosen

$$V_c = \int_{t-T}^{t} \left[u_{m1}^2(\sigma) + u_{s1}^2(\sigma) + u_{s2}^2(\sigma)\right] d\sigma. \tag{2}$$

By using (1), the first derivative of V_c can be obtained as

$$\dot{V}_c = u_{m1}^2(t) + u_{s1}^2(t) + u_{s2}^2(t) - z_{m1}^2(t) - z_{m2}^2(t) - z_{s1}^2(t). \tag{3}$$

From Fig. 4, the following relations can be inferred

$$\begin{aligned} u_{m1} &= w_{11}u_1 + w_{21}u_2 + w_{31}u_3 \\ z_{m1} &= w_{12}u_1 + w_{22}u_2 + w_{32}u_3 \\ z_{m2} &= w_{13}u_1 + w_{23}u_2 + w_{33}u_3 \\ z_{s1} &= v_{11}y_1 + v_{21}y_2 + v_{31}y_3 \\ u_{s1} &= v_{12}y_1 + v_{22}y_2 + v_{32}y_3 \\ u_{s2} &= v_{13}y_1 + v_{23}y_2 + v_{33}y_3. \end{aligned} \tag{4}$$

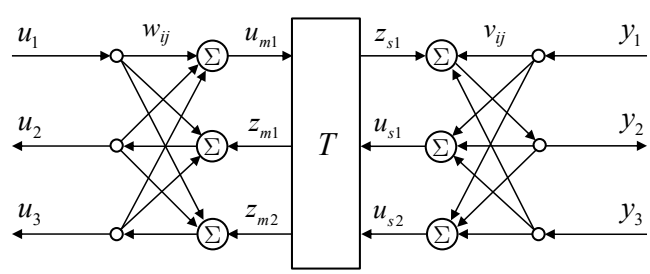

Fig. 4. The force-receptive scattering transformation.

Substituting (4) into (3) and rearranging the result yield

$$\begin{aligned}\dot{V}_c = &\left(w_{11}^2 - \sum_{k=2}^{3} w_{1k}^2\right) u_1^2 + \left(w_{21}^2 - \sum_{k=2}^{3} w_{2k}^2\right) u_2^2 \\ &+ \left(w_{31}^2 - \sum_{k=2}^{3} w_{3k}^2\right) u_3^2 + \left(\sum_{k=2}^{3} v_{1k}^2 - v_{11}^2\right) y_1^2 \\ &+ \left(\sum_{k=2}^{3} v_{2k}^2 - v_{21}^2\right) y_2^2 + \left(\sum_{k=2}^{3} v_{3k}^2 - v_{31}^2\right) y_3^2 \\ &+ 2\left(w_{11}w_{21} - \sum_{k=2}^{3} w_{1k}w_{2k}\right) u_1 u_2 \\ &+ 2\left(w_{11}w_{31} - \sum_{k=2}^{3} w_{1k}w_{3k}\right) u_1 u_3 \\ &+ 2\left(w_{21}w_{31} - \sum_{k=2}^{3} w_{2k}w_{3k}\right) u_2 u_3 \\ &+ 2\left(\sum_{k=2}^{3} v_{1k}v_{2k} - v_{11}v_{21}\right) y_1 y_2 \\ &+ 2\left(\sum_{k=2}^{3} v_{1k}v_{3k} - v_{11}v_{31}\right) y_1 y_3 \\ &+ 2\left(\sum_{k=2}^{3} v_{2k}v_{3k} - v_{21}v_{31}\right) y_2 y_3.\end{aligned} \tag{5}$$

Denoting $\boldsymbol{u} := [\delta_1 u_1 \;\; \delta_2 u_1 \;\; \delta_3 u_2 \;\; \Delta_1 y_1 \;\; \Delta_2 y_1 \;\; \Delta_3 y_2]^\top \in \mathbb{R}^6$ and $\boldsymbol{y} := 2\,[u_2 \;\; u_3 \;\; u_3 \;\; y_2 \;\; y_3 \;\; y_3]^\top \in \mathbb{R}^6$, where δ_i and Δ_i $(i \in \mathcal{N})$ are also to-be-determined parameters, which will be specified in section III-D. Let $\gamma_m \leq 0$ and $\gamma_s \leq 0$ be the desired level of excessive passivity at the port u_1 and y_1, respectively. Based on (5), the force-receptive scattering transformation passivates the communication channel with the passive input–output pair $\boldsymbol{u}$–$\boldsymbol{y}$ if the following constraints are fulfilled

$$\begin{aligned}&w_{11}^2 - \sum_{k=2}^{3} w_{1k}^2 = \gamma_m,\; w_{21}^2 - \sum_{k=2}^{3} w_{2k}^2 = 0,\; w_{31}^2 - \sum_{k=2}^{3} w_{3k}^2 = 0 \\ &\sum_{k=2}^{3} v_{1k}^2 - v_{11}^2 = \gamma_s,\; \sum_{k=2}^{3} v_{2k}^2 - v_{21}^2 = 0,\; \sum_{k=2}^{3} v_{3k}^2 - v_{31}^2 = 0, \\ &w_{11}w_{21} - \sum_{k=2}^{3} w_{1k}w_{2k} = \delta_1,\; w_{11}w_{31} - \sum_{k=2}^{3} w_{1k}w_{3k} = \delta_2, \\ &w_{21}w_{31} - \sum_{k=2}^{3} w_{2k}w_{3k} = \delta_3,\; \sum_{k=2}^{3} v_{1k}v_{2k} - v_{11}v_{21} = \Delta_1, \\ &\sum_{k=2}^{3} v_{1k}v_{3k} - v_{11}v_{31} = \Delta_2,\; \sum_{k=2}^{3} v_{2k}v_{3k} - v_{21}v_{31} = \Delta_3.\end{aligned} \tag{6}$$

When the neural weights w_{ij} and v_{ij} verify (6), we have

$$\dot{V}_c = \boldsymbol{u}^\top \boldsymbol{y} + \gamma_m u_1^2 + \gamma_s y_1^2. \tag{7}$$

Moreover, based on Fig. 4 and the relation (4), it is also required that $w_{22} \neq 0$, $w_{33} \neq 0$, and $v_{21} \neq 0$.

B. Steady-state transmission condition enforcement

Since the force-receptive scattering transformation is an autoencoder, the choice of the weights w_{ij} and v_{ij} should not allow direct data transmission, i.e., the condition $u_2(t) = y_2(t-T)$ and $u_3(t) = y_3(t-T)$ $\forall t \geq 0$ are prohibited. Nevertheless, one should preserve certain transmission criteria in the steady-state to ensure the correctness of data transfer when everything settles down, so that the teleoperation system is well-behaved. For this reason, the following steady-state transmission requirement is defined.

Definition 1 (Steady-state transmission requirement)**.** Considering the communication channel with the force-receptive scattering transformation as shown in Fig. 4. Assuming that all input/output signals of the system are in their steady states, i.e., the following limits

$$\lim_{t\to\infty} u_i(t) := \bar{u}_i,\; \lim_{t\to\infty} y_i(t) := \bar{y}_i \;\; (i \in \mathcal{N})$$

exist and are finite, and the condition $\bar{u}_1 = \bar{y}_1 = 0$ holds. Then, the communication channel is said to satisfy the steady-state transmission requirement if and only if

$$\bar{u}_i = \bar{y}_i \;\; (i = 2, 3)\,.$$

The following Theorem summarizes the constraints on the weights w_{ij} and v_{ij} to realize the above-mentioned steady-state transmission requirement.

Theorem 1. A necessary and sufficient condition for the force-receptive scattering transformation shown in Fig. 4 to fulfill the steady-state transmission requirement is

$$\begin{aligned}&w_{2i} = v_{2i},\; w_{3i} = v_{3i} \;\; (i \in \mathcal{N}) \\ &\operatorname{rank}(\boldsymbol{A}) = 2, \text{ where } \boldsymbol{A} = \begin{bmatrix} w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix}^\top.\end{aligned} \tag{8}$$

Proof. The proof is along the same lines as [11, Theorem 2]. □

C. Existence of solution

In this subsection, the solvability of the force-receptive scattering transformation will be discussed. To sum up the results of subsections III-A and III-B, the weights w_{ij} and v_{ij} must fulfill the following set of polynomial constraints

$$\begin{cases} \bullet\;\; w_{22} \neq 0,\; w_{33} \neq 0, \text{ and } v_{21} \neq 0 \\ \bullet\;\; \text{polynomial equations (6) (for passivity)} \\ \bullet\;\; \text{polynomial constr. (8) (for steady-state transmission).} \end{cases} \tag{9}$$

Excluding inequalities, the above set of polynomial constraints has 18 polynomial equations in a total of 26 free variables including $w_{ij}, v_{ij}, \delta_i, \Delta_i, \gamma_m$, and γ_s $(i, j \in \mathcal{N})$. Here, we are only interested in real solutions of (9). Since the number of equations and variables of (9) is relatively large, the computation of Gröbner basis—the main tool from algebraic geometry to study solutions of polynomial equations [14, p. 13]—is inefficient. Furthermore, it is next

to impossible to obtain closed-form expressions for all solutions. However, we argue that (9) has at least one real solution.

Theorem 2. Consider the system of polynomial constraints (9). If it is chosen that

$$\Delta_3 = -\delta_3 \neq 0,\ \frac{2\delta_1\delta_2}{\delta_3} - \gamma_m \geq 0,\ \frac{2\Delta_1\Delta_2}{\delta_3} + \gamma_s \geq 0, \quad (10)$$

then the force-receptive scattering transformation is solvable, i.e., the set of polynomial constraints (9) has at least one real solution of w_{ij} and v_{ij} $(i, j \in \mathcal{N})$.

Proof. The proof can be done by pointing out one instance of w_{ij} and v_{ij} that satisfies (9). Such an instance should be carefully selected to reduce the complexity in solving. After some trial and error, the following special solution is chosen

$$w_{22} = w_{21},\ w_{33} = -w_{31}.$$

Substituting this relation as well as the equations of (8) into (6) results in $\Delta_3 = -\delta_3$. Moreover, (9) can also be completely reduced to

$$\begin{aligned}
&w_{11}^2 - w_{12}^2 - w_{13}^2 = \gamma_m,\ w_{21}\left(w_{11} - w_{12}\right) = \delta_1,\\
&w_{31}\left(w_{11} + w_{13}\right) = \delta_2,\ w_{21}w_{31} = \delta_3,\ v_{12}^2 + v_{13}^2 - v_{11}^2 = \gamma_s,\\
&w_{21}\left(v_{12} - v_{11}\right) = \Delta_1,\ -w_{31}\left(v_{13} + v_{11}\right) = \Delta_2,\\
&w_{23} = 0,\ w_{32} = 0,\ v_{2i} = w_{2i},\ v_{3i} = w_{3i}\ (i \in \mathcal{N}),
\end{aligned} \quad (11)$$

in addition to the inequalities $w_{21} \neq 0$ and $w_{31} \neq 0$. Choosing $w_{21} := c \in \mathbb{R}\backslash\{0\}$, then from (11), we have

$$\begin{aligned}
&w_{12} = w_{11} - \delta_1 c^{-1},\ w_{13} = -w_{11} + \delta_2\delta_3^{-1}c,\ w_{31} = \delta_3 c^{-1}\\
&v_{12} = v_{11} + \Delta_1 c^{-1},\ v_{13} = -v_{11} - \Delta_2\delta_3^{-1}c,
\end{aligned}$$

where w_{11} and v_{11} are determined by

$$w_{11}^2 - 2\left(\frac{\delta_1}{c} + \frac{\delta_2 c}{\delta_3}\right)w_{11} + \left(\frac{\delta_1}{c}\right)^2 + \left(\frac{\delta_2 c}{\delta_3}\right)^2 + \gamma_m = 0,$$

$$v_{11}^2 + 2\left(\frac{\Delta_1}{c} + \frac{\Delta_2 c}{\delta_3}\right)v_{11} + \left(\frac{\Delta_1}{c}\right)^2 + \left(\frac{\Delta_2 c}{\delta_3}\right)^2 - \gamma_s = 0.$$

It is clear that, in order to have real solutions of w_{11} and v_{11}, the discriminants of the above quadratic equations need to be non-negative, namely that

$$\frac{2\delta_1\delta_2}{\delta_3} - \gamma_m \geq 0,\ \frac{2\Delta_1\Delta_2}{\delta_3} + \gamma_s \geq 0.$$

Moreover, it is required that $w_{31} \neq 0$, thus we need $\delta_3 \neq 0$ since $w_{31} = \delta_3/c$. This therefore completes the proof. □

In summary, the choice for δ_i, Δ_i, γ_m, and γ_s $(i \in \mathcal{N})$ should strictly adhere to (10) to guarantee the solvability of the force-receptive scattering transformation. Note also that (9) has more free variables than constraints, thus we can reasonably expect that multiple solutions of the neural weights w_{ij} and v_{ij} might exist. Some solutions might result in better control performance for the force-receptive scattering transformation than others. Therefore, in practice, numerical solvers can be employed to fully explore all possibilities.

D. Teleoperation using the force-receptive scattering transformation

In this subsection, we introduce a three-channel teleoperation system employing the force-receptive scattering transformation with environmental contact force feedback as shown in Fig. 5. Note that, the contact force (i.e., F_e) is transmitted independently from the kinematics information. In order to keep the subsequent analyses as simple as possible, one degree-of-freedom local and remote robots will be considered as similar to [1] and [11]. The dynamics of the local robot is given by

$$\ddot{x}_m = \tau_m + F_h, \quad (12)$$

where x_m, τ_m, and F_h denote the position, the applied motor force, and the applied human force of the local robot, respectively. Similarly, the remote robot is specified by

$$\ddot{x}_s = \tau_s - F_e, \quad (13)$$

where x_s, τ_s, and F_e represent the position, the applied motor force, and the environmental contact force of the remote robot, respectively.

Let $r_m := \dot{x}_m + \lambda x_m$ and $r_s := \dot{x}_s + \lambda x_s$ $(\lambda > 0)$. The local and remote robots are, respectively, controlled by

$$\tau_m = -\lambda\dot{x}_m + K_m\left(r_{md} - r_m\right) - F_{md} \quad (14)$$

$$\tau_s = -\lambda\dot{x}_s + K_s\left(r_{sd} - r_s\right) + \dot{r}_{sd} + F_e, \quad (15)$$

where $K_m, K_s > 0$ and local force feedback has been used on τ_s (see [15] and [16]). By substituting (14)–(15) into (12)–(13), the closed-loop systems of the local and remote sides are, respectively, given by

$$\dot{r}_m = -K_m\left(r_m - r_{md}\right) - F_{md} + F_h \quad (16)$$

$$\dot{r}_s - \dot{r}_{sd} = -K_s\left(r_s - r_{sd}\right). \quad (17)$$

From (17), the results $r_s - r_{sd} \to 0$ and $\dot{r}_s - \dot{r}_{sd} \to 0$ as $t \to +\infty$ can immediately be obtained. Moreover, $r_s - r_{sd} \in \mathcal{L}_\infty$ and $\dot{r}_s - \dot{r}_{sd} \in \mathcal{L}_\infty$. It is also noted that the following signal substitutions are performed on the general force-receptive scattering transformation shown in Fig. 4

$$\begin{aligned}
&u_1 = r_m - r_{md},\ u_2 = r_{md},\ u_3 = F_{md},\\
&y_1 = r_s - r_{sd},\ y_2 = r_{sd},\ y_3 = F_e.
\end{aligned} \quad (18)$$

The stability of the overall teleoperation system including the local side, the communication channel operated by the force-receptive scattering transformation, and the remote side can be analyzed by using the Lyapunov-Krasovskii functional $V = 0.5r_m^2 + 0.5\left(r_s - r_{sd}\right)^2 + 0.5V_c$, where V_c is given by (2), that is to say

$$V = \frac{1}{2}r_m^2 + \frac{1}{2}\left(r_s - r_{sd}\right)^2 + \frac{1}{2}\int_{t-T}^{t}\left(u_{m1}^2 + u_{s1}^2 + u_{s2}^2\right)d\sigma \quad (19)$$

By using (7) and (16)–(17), the first derivative of V is

$$\begin{aligned}
\dot{V} = &-K_m r_m\left(r_m - r_{md}\right) - F_{md}r_m + F_h r_m - K_s\left(r_s - r_{sd}\right)^2\\
&+ \delta_1\left(r_m - r_{md}\right)r_{md} + \delta_2\left(r_m - r_{md}\right)F_{md}\\
&+ \delta_3 F_{md}r_{md} + \Delta_1\left(r_s - r_{sd}\right)r_{sd} + \Delta_2\left(r_s - r_{sd}\right)F_e\\
&- \delta_3 F_e r_{sd} + \gamma_m\left(r_m - r_{md}\right)^2 + \gamma_s\left(r_s - r_{sd}\right)^2.
\end{aligned}$$

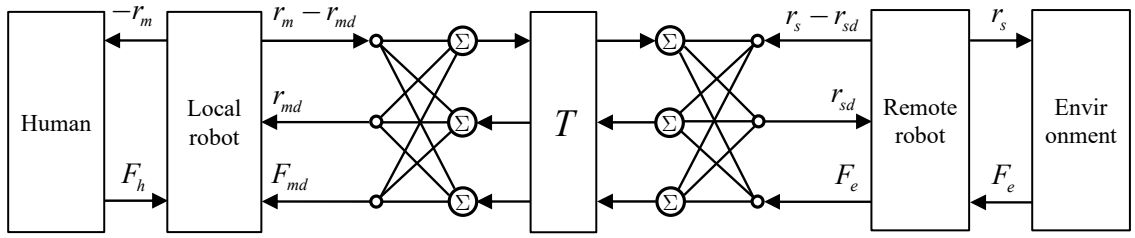

Fig. 5. A three-channel teleoperation architecture using the force-receptive scattering transformation.

Rearranging the previous equation yields

$$\begin{aligned}\dot{V} &= (r_m - r_{md})(\delta_1 r_{md} - K_m r_m) \\ &\quad + [(\delta_2 - 1) r_m - (\delta_2 - \delta_3) r_{md}] F_{md} + F_h r_m \\ &\quad + \Delta_1 (r_s - r_{sd}) r_{sd} - (\Delta_2 + \delta_3) F_e r_{sd} + \Delta_2 F_e r_s \\ &\quad - K_s (r_s - r_{sd})^2 + \gamma_m (r_m - r_{md})^2 + \gamma_s (r_s - r_{sd})^2 .\end{aligned} \tag{20}$$

Based on (20), one can choose

$$\begin{aligned}&\delta_1 = K_m,\ \delta_2 = 1, \delta_3 = \delta_2 = 1, \\ &\Delta_1 = 0,\ \Delta_2 = \Delta_3 = -\delta_3 = -1,\ \gamma_m = \gamma_s = 0,\end{aligned} \tag{21}$$

to obtain the following result

$$\dot{V} = -K_m (r_m - r_{md})^2 - K_s (r_s - r_{sd})^2 + F_h r_m - F_e r_s. \tag{22}$$

Remark 1. It can straightforwardly be checked that (21) satisfies the condition (10), and thus, the solvability of the force-receptive scattering transformation is always guaranteed while the validity of (22) is maintained.

IV. Stability analysis

Before conducting the stability analysis for the overall system, the bounded-input-bounded-output stability of the force-receptive scattering transformation will be given in the following Theorem. For the sake of convenience, let us denote $\omega_4 := (w_{21}(1+\omega_2))^{-1}(\omega_3 - w_{21}\omega_2)$, where $\omega_2 := \omega_1 (w_{13}w_{32} - w_{12}w_{33})$, $\omega_1 := (w_{22}w_{33} - w_{23}w_{32})^{-1}$, and $\omega_3 := -w_{11} + w_{31}\omega_1 (w_{13}w_{22} - w_{12}w_{23})$.

Theorem 3. Consider the force-receptive scattering transformation as shown in Fig. 4, whose structure is governed by the relations (1) and (4) subject to the signal substitution (18). Assume that the neural weights w_{ij} and v_{ij} $(i, j \in \mathcal{N})$ are determined by (9) (with the variable selection (21)) such that $|\varepsilon| < 1$ for all $\zeta \geq 0$, where the coefficient ε is defined by

$$\varepsilon = \frac{1 + w_{21}^{-1} w_{31} \zeta + \omega_4}{1 + w_{21}^{-1} w_{31} \zeta}. \tag{23}$$

If $r_s(t)$ is supposed to be bounded for all $t < 0$, then whenever $r_m \in \mathcal{L}_\infty$ and $r_s - r_{sd} \in \mathcal{L}_\infty$, we have $r_s \in \mathcal{L}_\infty$. Moreover, the outputs of the force-receptive scattering transformation including r_{md}, r_{sd}, and F_{md} also belong to $\mathcal{L}_\infty$.

The proof of Theorem 3 is omitted here due to lack of space. Generally, the fulfillment of the requirement $|\varepsilon| < 1$ $\forall \zeta \geq 0$ depends on the values of the resulting neural weights w_{ij} and v_{ij} of the force-receptive scattering transformation. However, we can show that there is always one instance of w_{ij} and v_{ij} that makes $|\varepsilon| < 1$ for all $\zeta \geq 0$. Continuing the proof of Theorem 2 on the solvability of the force-receptive scattering transformation, we have $\omega_3 = -w_{21}$ if the variable selection (21) is kept. Since $\omega_4 = (w_{21}(1+\omega_2))^{-1}(\omega_3 - w_{21}\omega_2)$, it is true that $\omega_4 = -1$. Therefore, $\varepsilon = \zeta w_{21}^{-1} w_{31} \left(1 + w_{21}^{-1} w_{31} \zeta\right)^{-1} = \zeta c^{-2} \left(1 + \zeta c^{-2}\right)^{-1}$, where the results $w_{21} = c \neq 0$ and $w_{31} = \delta_3 / c = 1/c$ have been used. It is clear that, in this case, $|\varepsilon| < 1$ for all $\zeta \geq 0$.

The stability of the closed-loop teleoperation system utilizing the proposed force-receptive scattering transformation is summarized in the next Theorem.

Theorem 4. Consider the teleoperation system as shown in Fig. 5, where the local and remote robots (12)–(13) are controlled by the control laws (14)–(15) and the communication channel is orchestrated by the force-receptive scattering transformation. Suppose that all assumptions of Theorem 3 remain valid. It is also assumed that $(F_h, -r_m)$ and (F_e, r_s) define passive maps such that $r_m \in \mathcal{L}_\infty$ implies $F_h \in \mathcal{L}_\infty$ and $r_s \in \mathcal{L}_\infty$ implies $F_e \in \mathcal{L}_\infty$. Then, all signals of the closed-loop teleoperation system are uniformly bounded.

Proof. Because $(F_h, -r_m)$ and (F_e, r_s) define passive maps, it is true that

$$\int_0^t (-F_h r_m + F_e r_s)\, d\sigma \geq 0.$$

Define a new Lyapunov-Krasovskii functional V_1 as follows

$$V_1(t) = V(t) + \int_0^t (-F_h r_m + F_e r_s)\, d\sigma,$$

where $V(t)$ is given in (19). By using (22), the first derivative of V_1 can be computed as

$$\dot{V}_1 = -K_m (r_m - r_{md})^2 - K_s (r_s - r_{sd})^2 \leq 0.$$

From the above differential inequality, it is true that $V_1(t) \leq V_1(0) \in \mathbb{R}^+$ for all $t \geq 0$, and thus $V_1 \in \mathcal{L}_\infty$. As a consequence, $r_m \in \mathcal{L}_\infty$ and $r_s - r_{sd} \in \mathcal{L}_\infty$. Applying these results on Theorem 3, the conclusion $r_s, r_{md}, r_{sd}, F_{md} \in \mathcal{L}_\infty$ can also be obtained. Since $r_m, r_s \in \mathcal{L}_\infty$ and $r_i = \dot{x}_i + \lambda x_i$ $(i = m, s)$, the results $\dot{x}_i, x_i \in \mathcal{L}_\infty$ hold, i.e., the positions and velocities of both local and remote robots are bounded.

Since $r_m, r_s \in \mathcal{L}_\infty$, we have $F_h, F_e \in \mathcal{L}_\infty$ by assumption. According to (16), the result $\dot{r}_m \in \mathcal{L}_\infty$ is obtained. Based on

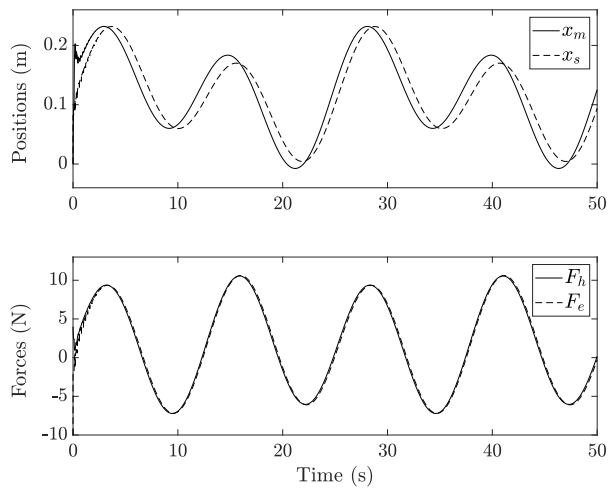

Fig. 6. Simulation result of the force-receptive scattering transformation-based teleoperation system.

the structure of the force-receptive scattering transformation, it can also be proved that $\dot{r}_m \in \mathcal{L}_\infty$ implies $\dot{r}_s \in \mathcal{L}_\infty$. Thus, $\dot{r}_{sd}$ also belongs to $\mathcal{L}_\infty$ because $\dot{r}_s - \dot{r}_{sd} \in \mathcal{L}_\infty$ (see section III-D). By gathering all of the above-mentioned information, one can conclude that $\tau_m, \tau_s \in \mathcal{L}_\infty$ by using (14)–(15), i.e., the applied motor forces of the two robots are uniformly bounded. This therefore completes the proof. □

V. Simulation result

The simulation is carried out to verify the efficacy of the proposed force-receptive scattering transformation-based teleoperation system. It is assumed that the remote robot is in contact with a dynamic object that has a moderate stiffness. For this reason, the environmental force is modeled by $F_e(t) = K_e\left(r_s(t) - d_s\right) + f_e^\star(t)$, where $K_e = 100$ N/m, $d_s = 0.1$ m, and $f_e^\star(t) = -5\sin(0.25t)$ if $t < 50$ s and $f_e^\star(t) = 0$ for $t \geq 50$ s. The human force is specified by $F_h(t) = -K_h\left(r_m(t) - d_m\right) + f_h^\star(t)$, where $K_h = 20$ N/m, $d_m = 0.2$ m, and $f_h^\star(t) = 10\sin(0.5t)$ if $t < 50$ s and $f_h^\star(t) = 0$ for $t \geq 50$ s. The magnitude of the time delay is $T = 0.1$ second. The control gains are selected as $K_m = K_s = 40$ and $\lambda = 1$. By employing the chosen values of K_m and K_s, the weights of the force-receptive scattering transformation can be computed by MATLAB as $w_{11} = -4.98$, $w_{12} = -0.41$, $w_{13} = -4.96$, $w_{21} = -4.11$, $w_{22} = 1.53$, $w_{23} = 3.81$, $w_{31} = -0.19$, $w_{32} = -0.19$, $w_{33} = 0.02$, $v_{11} = -4.11$, $v_{12} = 1.53$, $v_{13} = 3.81$, $v_{21} = -4.11$, $v_{22} = 1.53$, $v_{23} = 3.81$, $v_{31} = -0.19$, $v_{32} = -0.19$, and $v_{33} = 0.02$. The simulation result is shown in Fig. 6. It can be observed that both positions and forces of the two robots are closely synchronized, providing a high level of teleoperation transparency for the human operator.

Remark 2. In Theorem 4, the stability of the closed-loop teleoperation system is investigated in the case that both human and remote environment are assumed to be passive. The stability analysis for the case when the human and environmental forces contain non-passive components (as presented in the above simulation) was already successfully conducted. Nevertheless, such a result is not included here due to lack of space, and it will be documented in the upcoming report. Moreover, a detailed comparison between the proposed force-receptive scattering transformation-based teleoperation architecture with the existing frameworks will be provided in the future.

VI. Conclusions and future work

This paper proposed a force-receptive scattering transformation scheme which directly grew out from the observation that the classical scattering transformation is a neural network. A bridge has been introduced to connect the scattering transformation theory with the body of the neural network field. Our future work is twofold. First, the case of time-varying delay will be considered. Second, we will look into different neural network architectures and also adopt their learning features to obtain even better transparency with less restrictive stability results when dealing with more general classes of teleoperation architectures and/or time delay types.

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